



NICE3000new
User Manual



NICE3000new
Electrical Drawing

Version NO. V3.01

Elevator control system electric appliance schematic drawing

NICE3000new

Drawing:

Check:

Date:

Standard of NICE3000new electric wiring diagram :

1. Power level naming rules:

380V power supply in front of the main air switch three-phase five-wire named L1, L2, L3, N, PE, in the main air switch after the three-phase five-wire naming according to R, S, T, N, PE principle.

220V Illumination Power supply is named L1, N in front of the air switch of the power cabinet, and named 501,502 in the rear stage of the air switch. After that, the named fire line is graded according to 501 rule, and the range is between 503-510.

The primary control transformer is based on the R,S,T rule. The secondary naming rules of the control transformer are AC 220V:101, 102, AC110V: 02,01and DC110V: 22,21.

The input terminals of 24V switching power supply are 101 and 102, and the output terminals are 301 and 302.

2. Line number naming rules:

The controller 24V input signal correlation line uses the code corresponding to the signal as the line number, 01-18,107-112 as AC110V correlation line; and 101-102 as AC220V related circuit for control, 501-510 as AC220V lighting circuit related circuit,

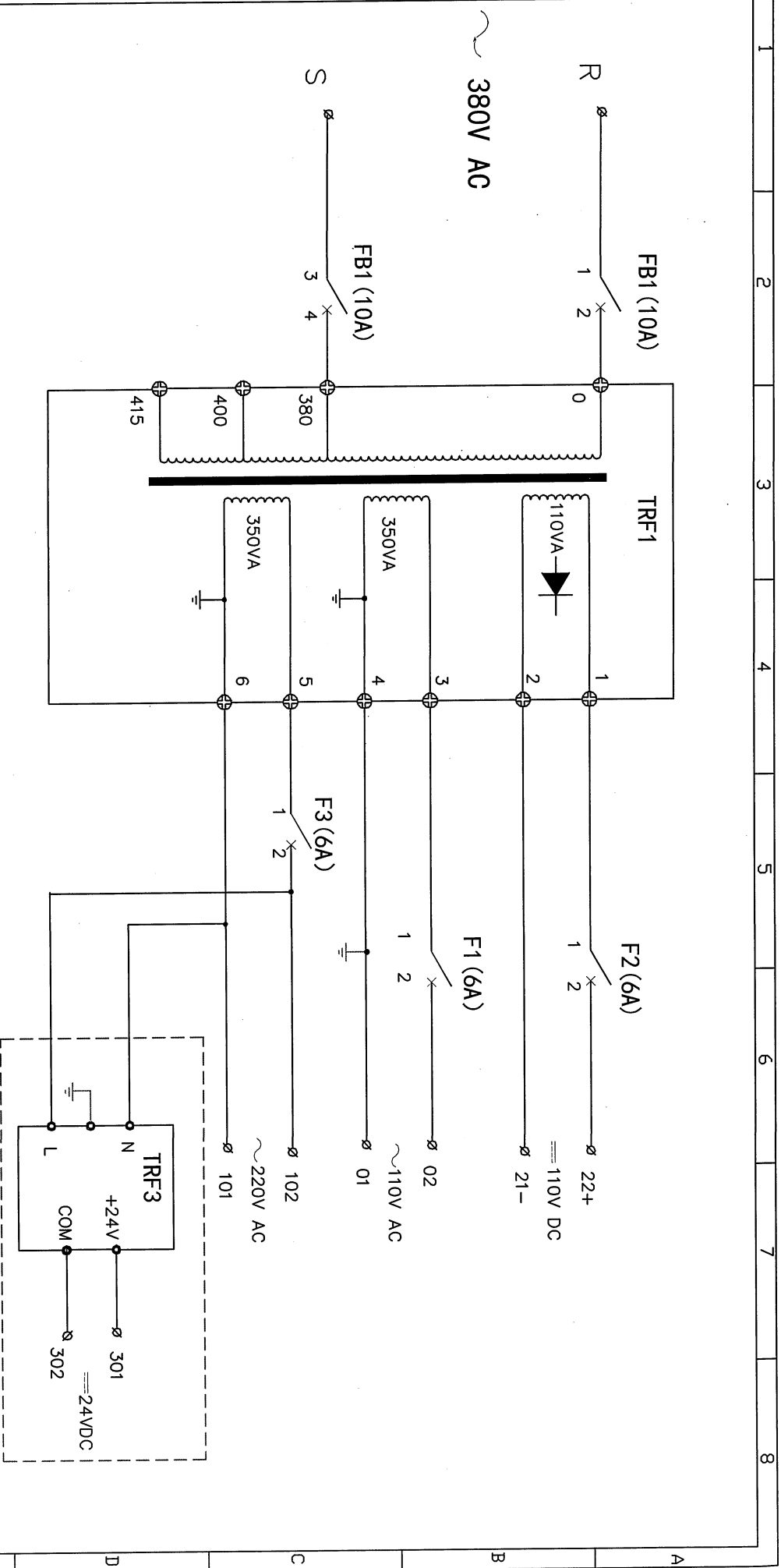
3. Switch and contact rules:

All switches, contactor contacts in accordance with the upper closed and down open, left open and right closed habit.

Drawing List		
NO.	Name	Notes
1	Standard and drawing list	
2	Electrical code description	
3	Power supply circuit	
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5	Safety and braker circuit	
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A	Code	Location	Name	Code	Location	Name	Code	Location	Name	Code	Location	Name	Code	Location	Name
	IBFS	HTW	Counterweight buffer switch	DL.S2	HTW	2nd down running slowdown switch	LSD	HTW	Down limit switch			Blackout emergency relay			
	2BFS	HTW	Car buffer switch	DL.S3	HTW	3rd down running slowdown switch	LSU	HTW	Up limit switch			Terminal row A			
	AB	CAR	Alarm bell	DM	CAR	Door motor	LWD	CAR	Weighing device	TA	CTR	Terminal row B			
	ACB	CAR	Attendant commutation button	DOB1	CAR	Door open button(front door)	LWO	CAR	Over load switch	TB	CTR	Top of car inspection switch			
	AP	CAR	Light curtain for front door	DOB2	CAR	Door open button(rear door)	LWX	CAR	Full load switch	TCI	CAR	Top of car inspection common button			
	AP1	CAR	Light curtain for back door	DS	HTW	Door lock switch	MCB	CTR	Main control board	TCIB	CAR	Top of car inspection down button			
	ARD	CTR	Blackout emergency flat layer devices	EEC	CAR	Emergency exit contact	MES	CTR	Machine room emergency stop button	TCID	CAR	Top of car inspection up button			
	ALB	CAR	Alarm bell button	EIS	CTR	Emergency inspection switch	MES-2	MR	Machine room emergency stop switch	TCIU	CAR	Top of car emergency stop switch			
	ATS	CAR	Attendant key switch	F2	CTR	DC110V power switch	MT	MR	Motor overheat protect switch	TES	CAR	Top of car inspection light			
	BRM	CTR	Brake resistance	F1	CTR	AC110V power switch	M	MR	Motor	TOCL	CAR	Top of car inspection light switch			
				F3	CTR	AC220V power switch	NSB	CAR	Nonstop switch	TOCLS	CAR	Top of car inspection light switch			
	BUZ	CAR	Buzzer	F9	CTR	UPS power switch	OS	HTW	Over speed governor switch	TRF1	CTR	Transformer NO.1			
	BM	MR	Brake	F	CAR	Car fan	PES1	HTW	Pit emergency switch 1	TRF2	CTR	Transformer NO.2			
	BRD	HTW	Governor remote action	FJR	CTR	Fire monitoring	PES2	HTW	Pit emergency switch 2	TRF3	CTR	Transformer NO.3			
	BDT	HTW	Governor remote release	FJRS1	HTW	Fir switch (backing floor)	PS	CAR	Up levelling switch	TUR1	CAR	Top of car utility receptacle AC220V			
				FJRS2	CAR	Fir switch (COP)	PX	CAR	Down levelling switch	TUR2	CAR	Top of car utility receptacle AC36V			
	CZA	CTR	Loose brake button	FJSD	HTW	Down final limit switch	PG	MR	Machine encoder						
	CBn	CAR	Car button n	FJSU	HTW	Up final limit switch				UDB	CTR	Up direction inspection button			
	CCB	CAR	Car contontrol board	FS	CAR	Car fan switch	PLB	CTR	Door lock control board	ULS1	HTW	1st top safety clearance limit switch			
	CES	CAR	Car emergency switch	FX	CAR	Shorting motor stator contactor	PLI	HTW	Pit light	ULS2	HTW	2nd top safety clearance limit switch			
	CEI	CAR	Car fluorescence light	GS	CAR	Gate switch	PLS1	HTW	Pit light switch 1	ULS3	HTW	3rd top safety clearance limit switch			
	CIS	CTR	Controller inspection switch	GTS	HTW	Governor tension switch	PLS2	HTW	Pit light switch 2	UPC	CTR	Uninterruptible power supply			
	CIU	CAR	Car light	HCB	HTW	Hall call and display board	PUR1	HTW	Pit socket AC220V						
	CUS	CAR	Car light switch	INTB	CAR	Interphone button	PUR2	HTW	Pit socket AC36V	XPX	HTW	Down re-level switch			
	CHM	CAR	Electric broadcaster	ISS	CAR	Independent running				XF	CTR	Fir switch signal			
	CTB	CAR	Car top board	IPS	CAR	Emergency power supply DC12V	Q1	CTR	Main power switch for machine room-less	XSQ	CTR	Governor test switch			
	CIU	CAR	Car inspection up button	KK	CTR	Running contactor	RGS	MR	Rope gripper switch	ZHKG	CTR	Turning switch			
	CID	CAR	Car inspection down button	KB	CTR	Brake contactor				ZL4	CTR	Rectifier			
	CINS	CAR	Car inspection switch	KD	CTR	Controller light	SDG	HTW	Levelling safety switch						
	DBR	CTR	Brake resistance	KZG	CTR	Controller power socket	SPS	HTW	Up re-level switch						
	DCB1	CAR	Door close button (front door)				SOS	CTR	Safety operated switch						
	DCB2	CAR	Door close button (rear door)	LHh	HTW	Holstway lighting	SK	HTW	Door open limit(front door)	CAR	NULL	Car			
	DDB	CTR	Down direction inspection button	LHS	HTW	Holstway lighting power switch	ISK	HTW	Door open limit(rear door)	CTR	NULL	Controller			
	DDCB	CAR	Door delay close button	LPS	CTR	Lighting power switch	SG	CAR	Door close limit(front door)	MR	NULL	Motor			
	DL.S1	HTW	1st down running slowdown switch	LPT	CAR	Arrival gong	ISG	CAR	Door close limit(rear door)	HTW	NULL	Holstway			

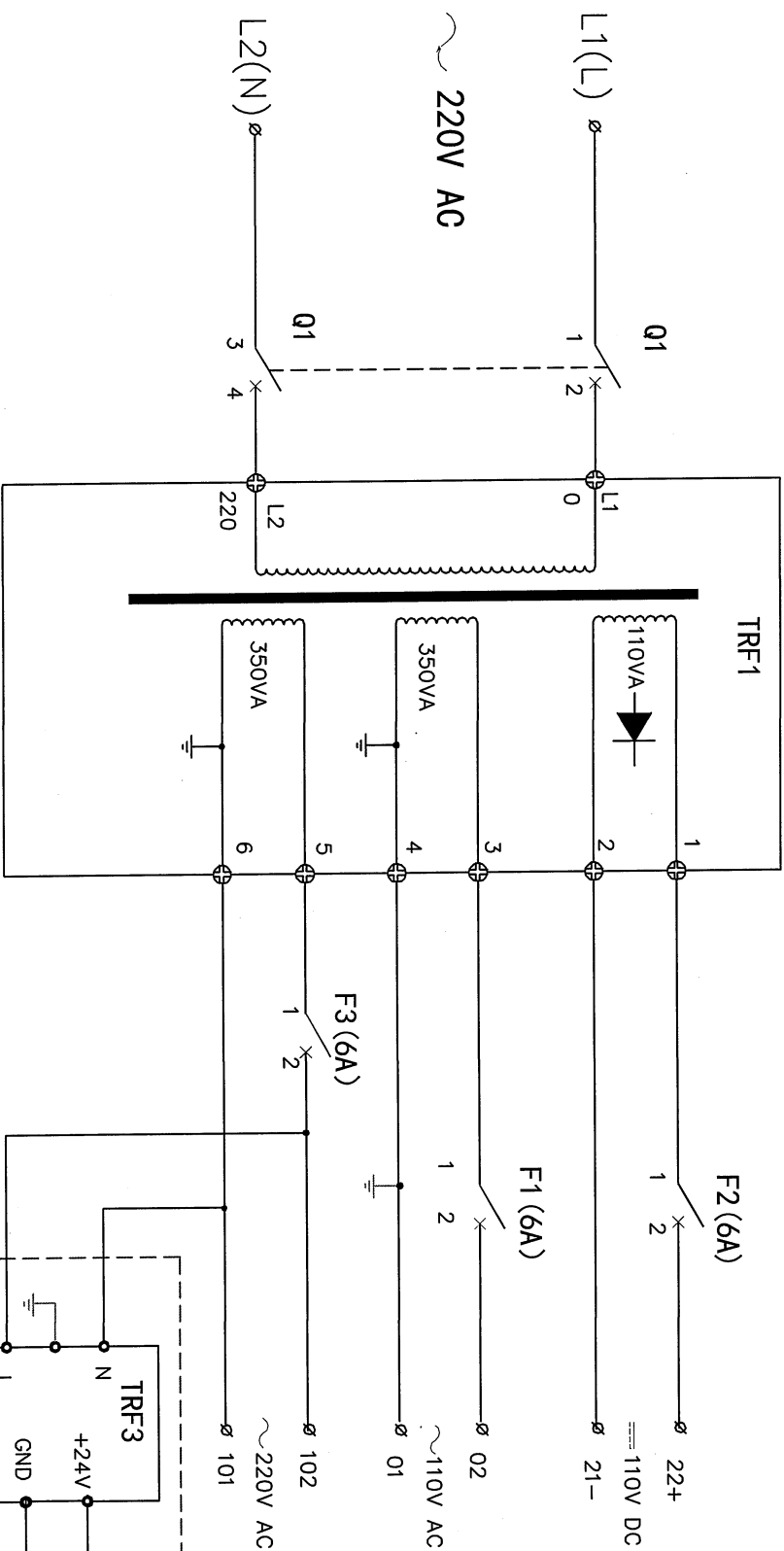
Design		Standard		Electrical code description	
Collate		Drawing NO.			
Check		Versions	V3.01	Page 2	NICE3000new
Craft		Date		Total 20	



Note:

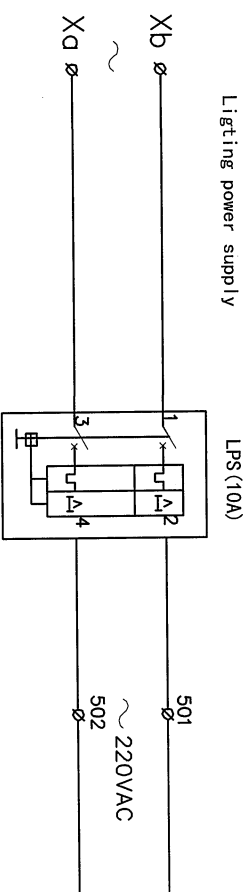
- 1, Choose the transformer TRF3 output current is 5A when the totality floors ≤ 10 , and it's 10A when totality floor > 10 .
- 2, This power supply circuit can be used when have UPS.

Design	Standard	Power supply circuit AC380V	
Collate	Drawing NO.	Page 3A	
Check	Version	V3.01	
Craft	Date	Total 20	NICE3000new

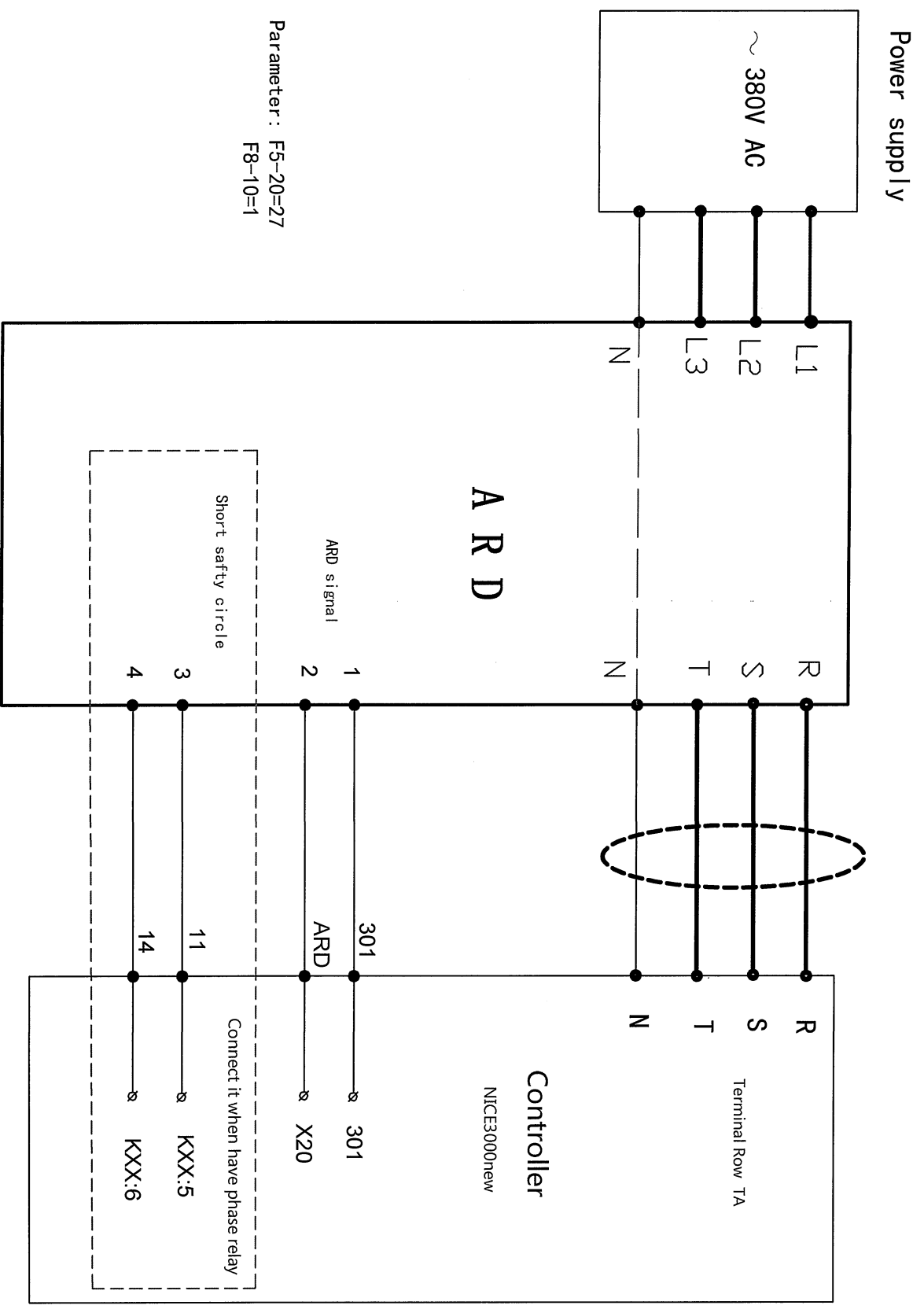


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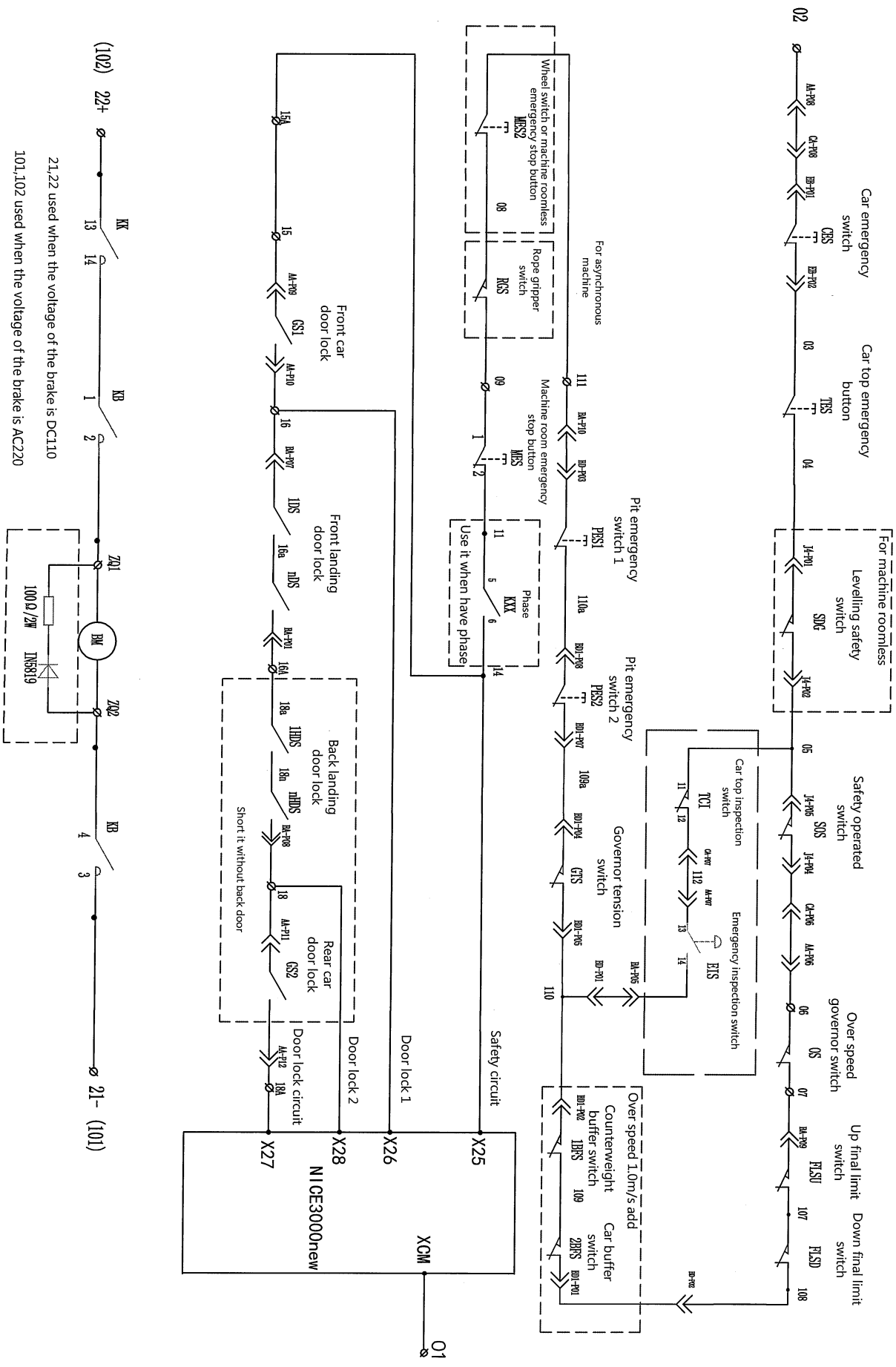
- 1, Choose the transformer TRF3 output current is 5A when the totality floors ≤ 10 , and it's 10A when totality floor > 10 .
- 2, This power supply circuit can be used when have UPS.
- 3, When input power 3 phase AC220V connect L1, L2, L3, When input power 2 phase AC220V connect L, N.



Design	Standard	Power supply circuit AC220V	
Collate	Drawing NO.		
Check	Versions	V3.01	Page 38
Craft	Date	Total 20	NICE3000new

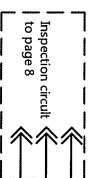
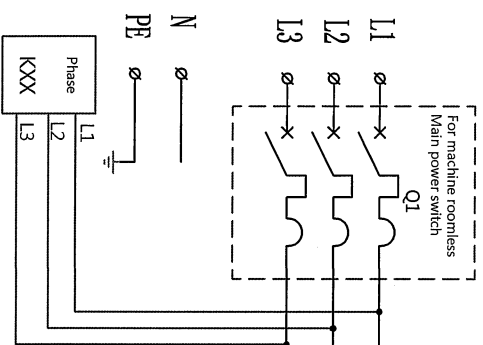


Design	Standard	ARD circuit	
Collate	Drawing NO.		
Check	V3.01	Page 4	NICE3000new
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			Safety and braker circuit	
Design		Standard		
		Drawing		
Collate		NO.		
Check		Versions	V3.01	Page 5
Craft		Date		
			Total 20	NICE3000new

Notes: 1. The 3-phase 5 wires of the power provided by user power supply box.
2. 37KW and the above apply brake unit.

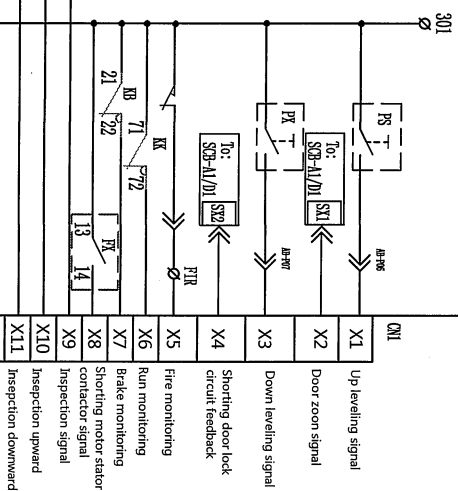


Model	Power	Brake Resistor
NICE-L-C-4005	5.5KW	1600W, 100Ω
NICE-L-C-4007	7.5KW	2500W, 70Ω
NICE-L-C-4011	11KW	3500W, 46Ω
NICE-L-C-4015	15KW	4500W, 40Ω
NICE-L-C-4018	18.5KW	5500W, 30Ω
NICE-L-C-4022	22KW	6500W, 23Ω
NICE-L-C-4030	30KW	9000W, 18Ω

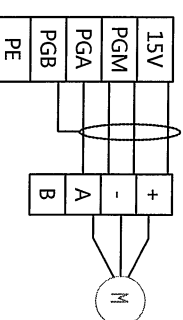
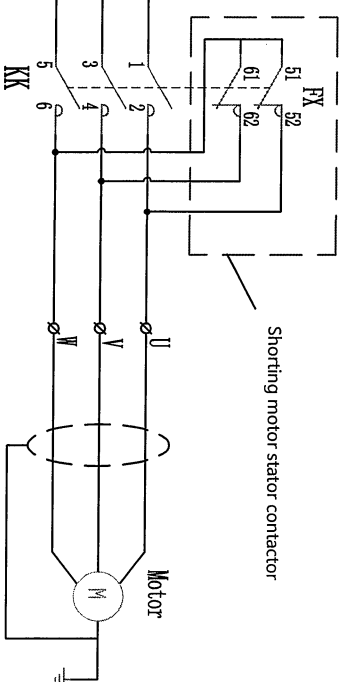
Brake resistance

DBR1-N

NICE3000new



Shorting motor stator contactor



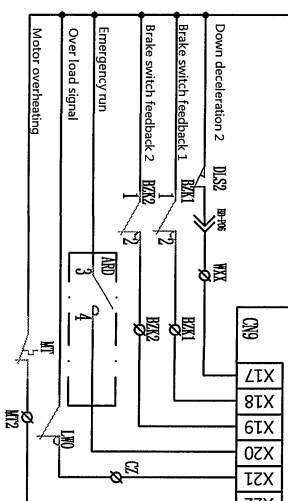
Encoder of asynchronous motor

Notes :
1. The synchronous motor use the PG card PG-E, and we choose SIN/COS encoder.

2. The asynchronous motor use the PG card PG-A2, and we choose push-pull encoder or open-collector incremental encoder.

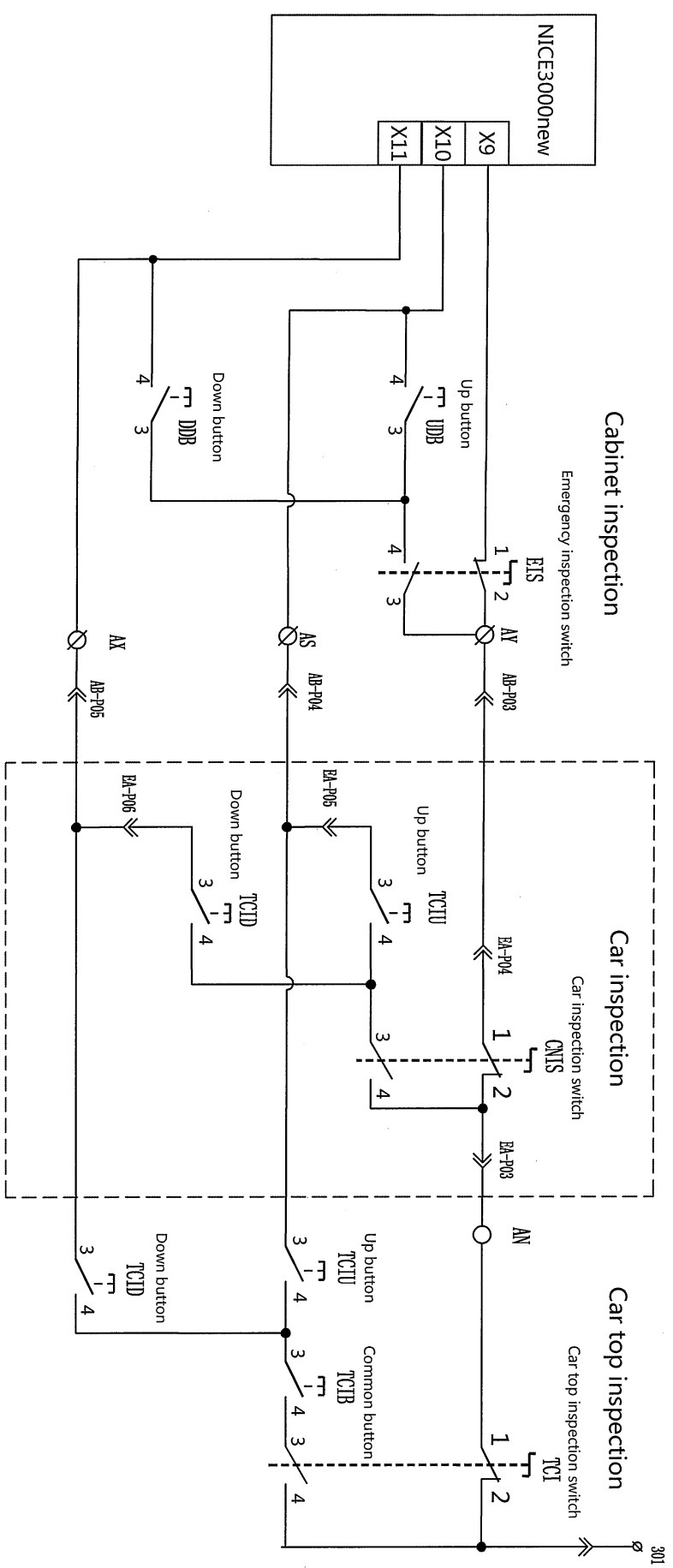
3. The shielding wire of the encode should connect at one side.

4. Soft limit function : F6-40 bit1=1

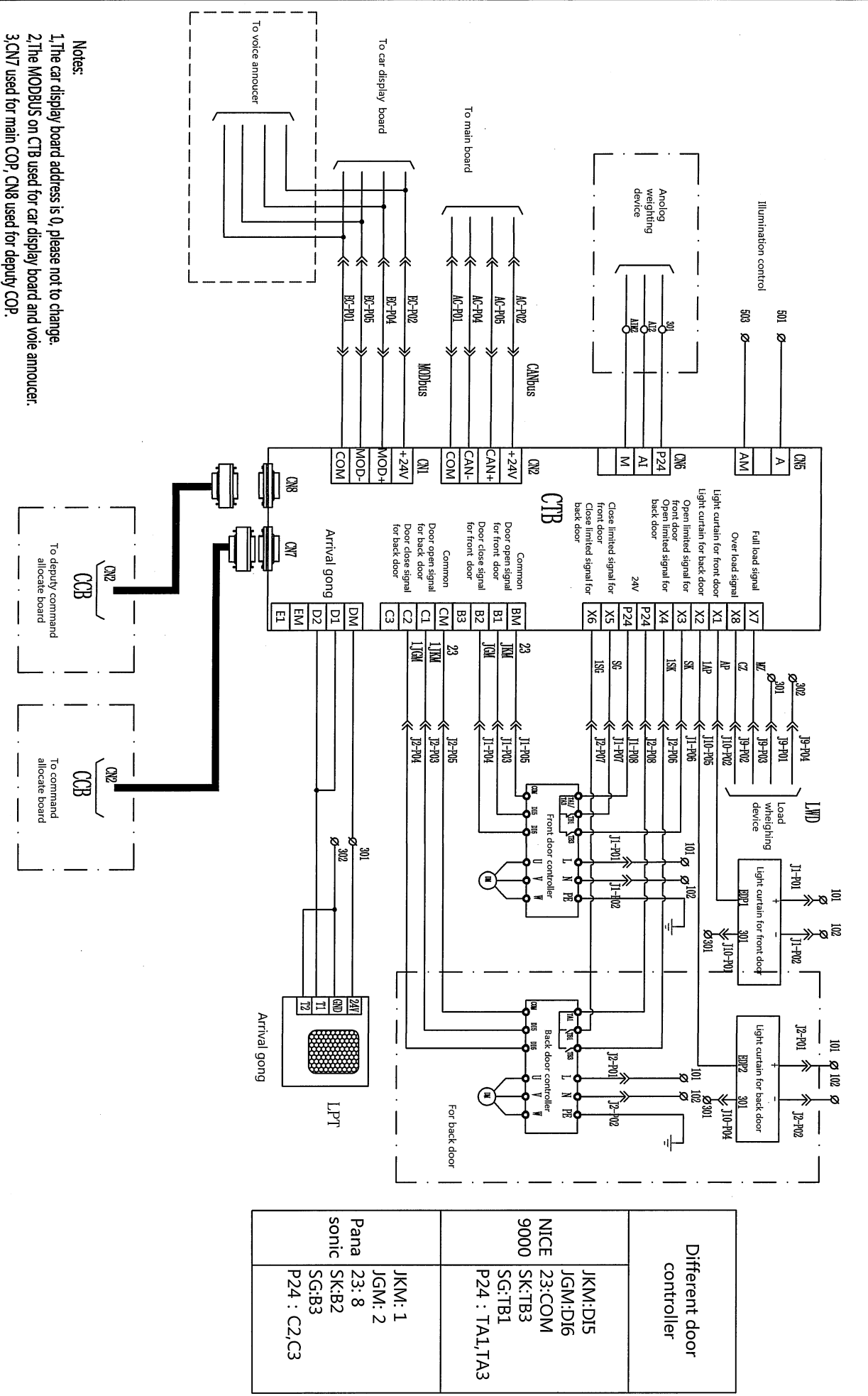


Design	Standard	Main controller circuit
Collate	Drawing	
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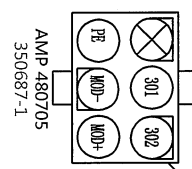
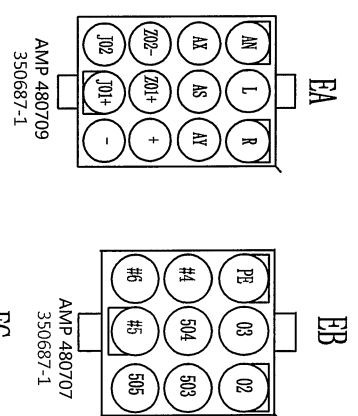
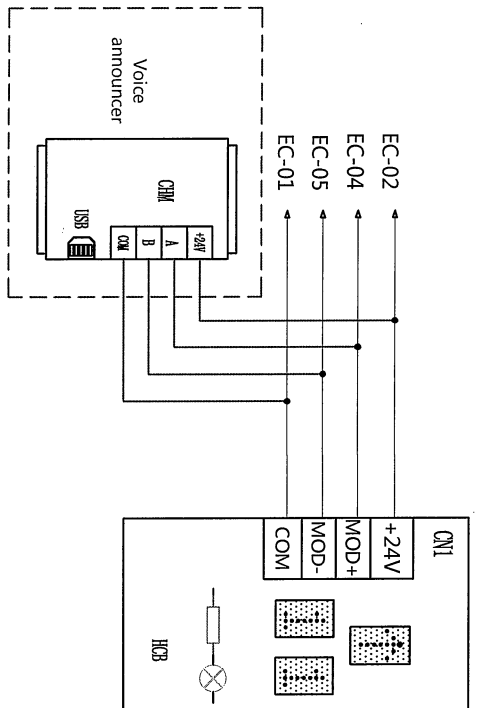
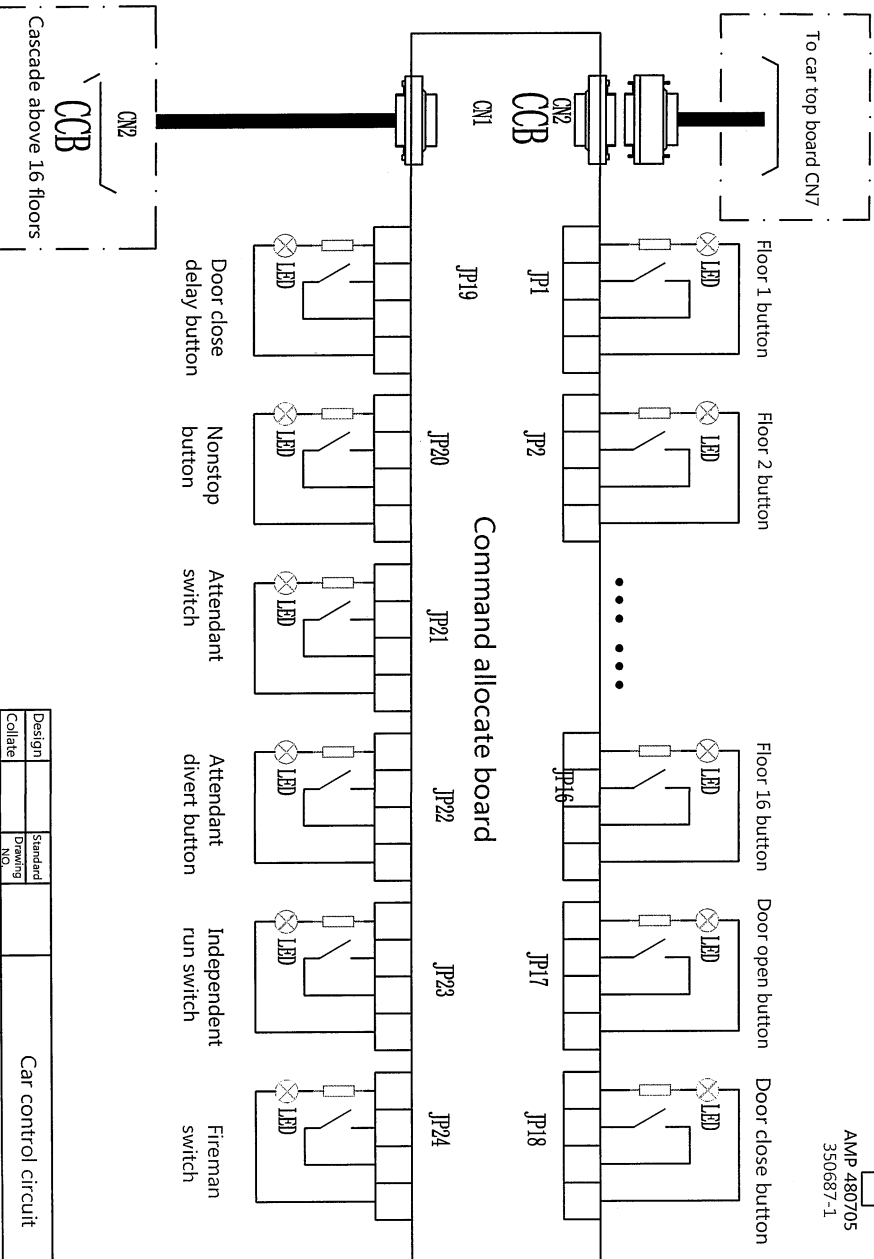
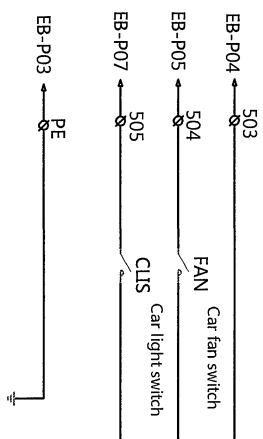
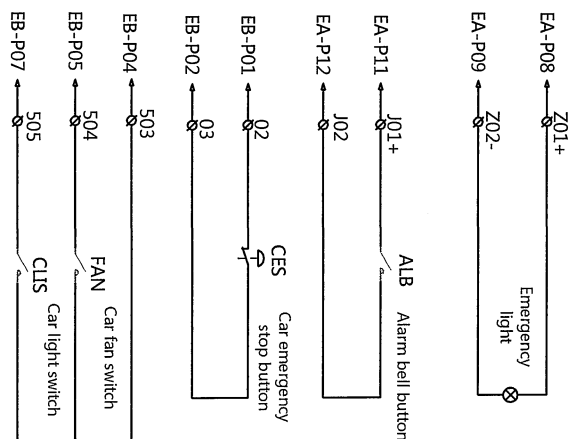
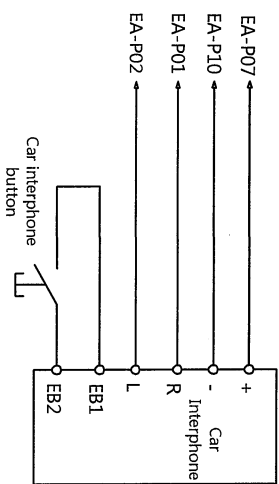
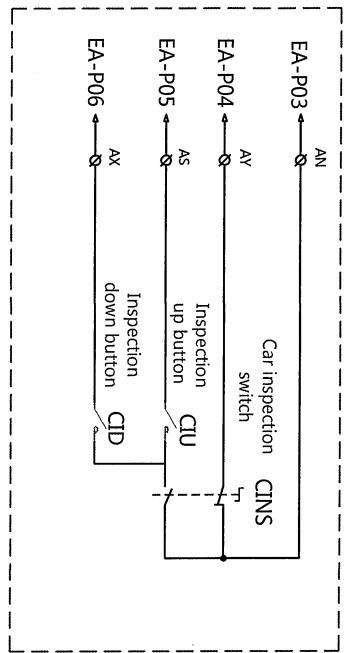


Design	Standard	Inspection circuit	
Collate	Drawing No.		
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- Notes:
1. The car display board address is 0, please not to change.
 2. The MODBUS on CTB used for car display board and voice announcer.
 3. CN7 used for main COP, CN8 used for deputy COP.

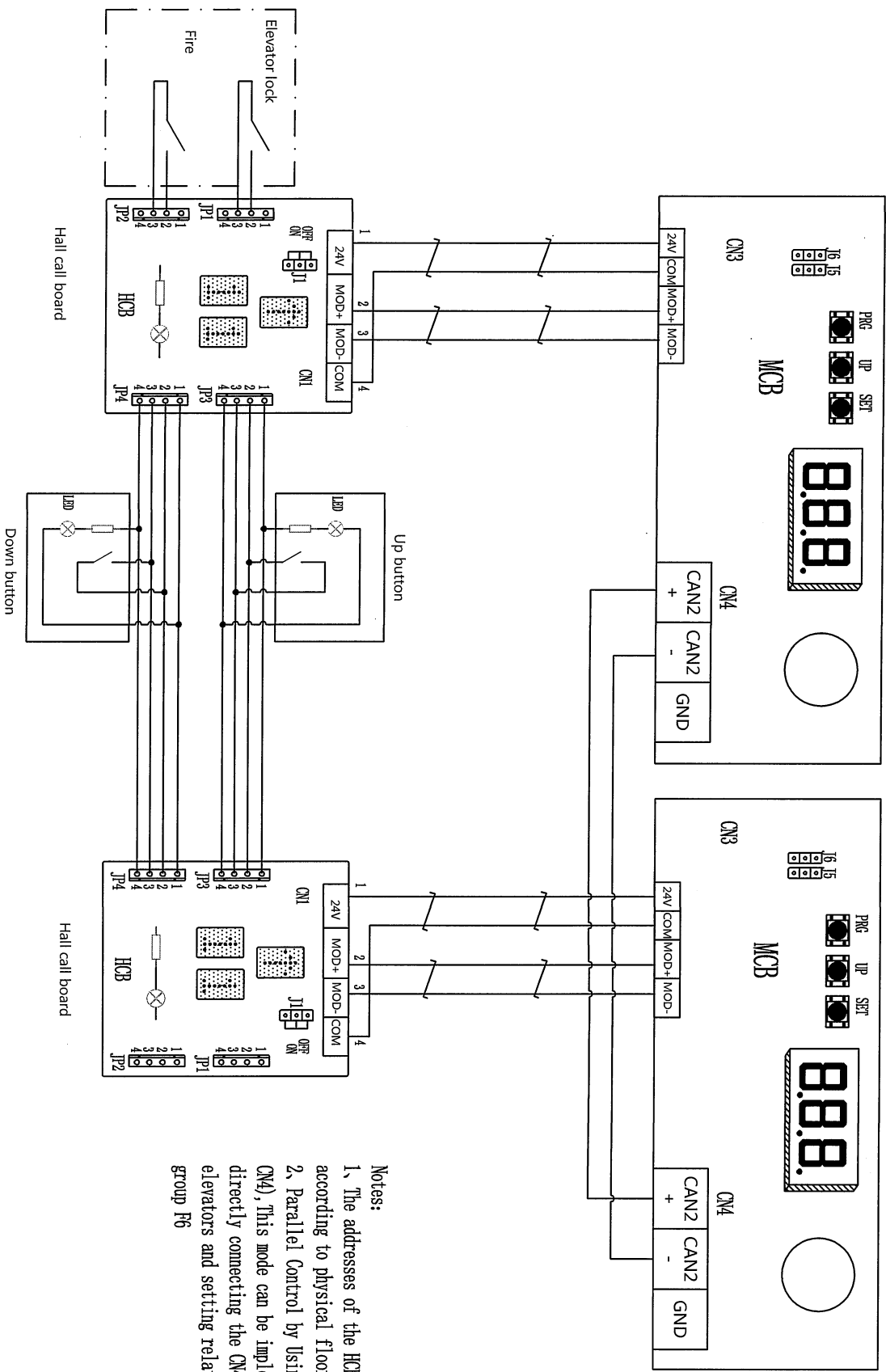
Design	Standard	Page 10	Car top control circuit
Collate	Drawing NO.	V3.01	
Check	Versions	Page 10	
Craft	Date	Total 20	
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Design	Standard	Car control circuit	
Collate	Drawing NO.	Page 11	NICE3000new
Check	Versions	V3.01	
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Lift 1

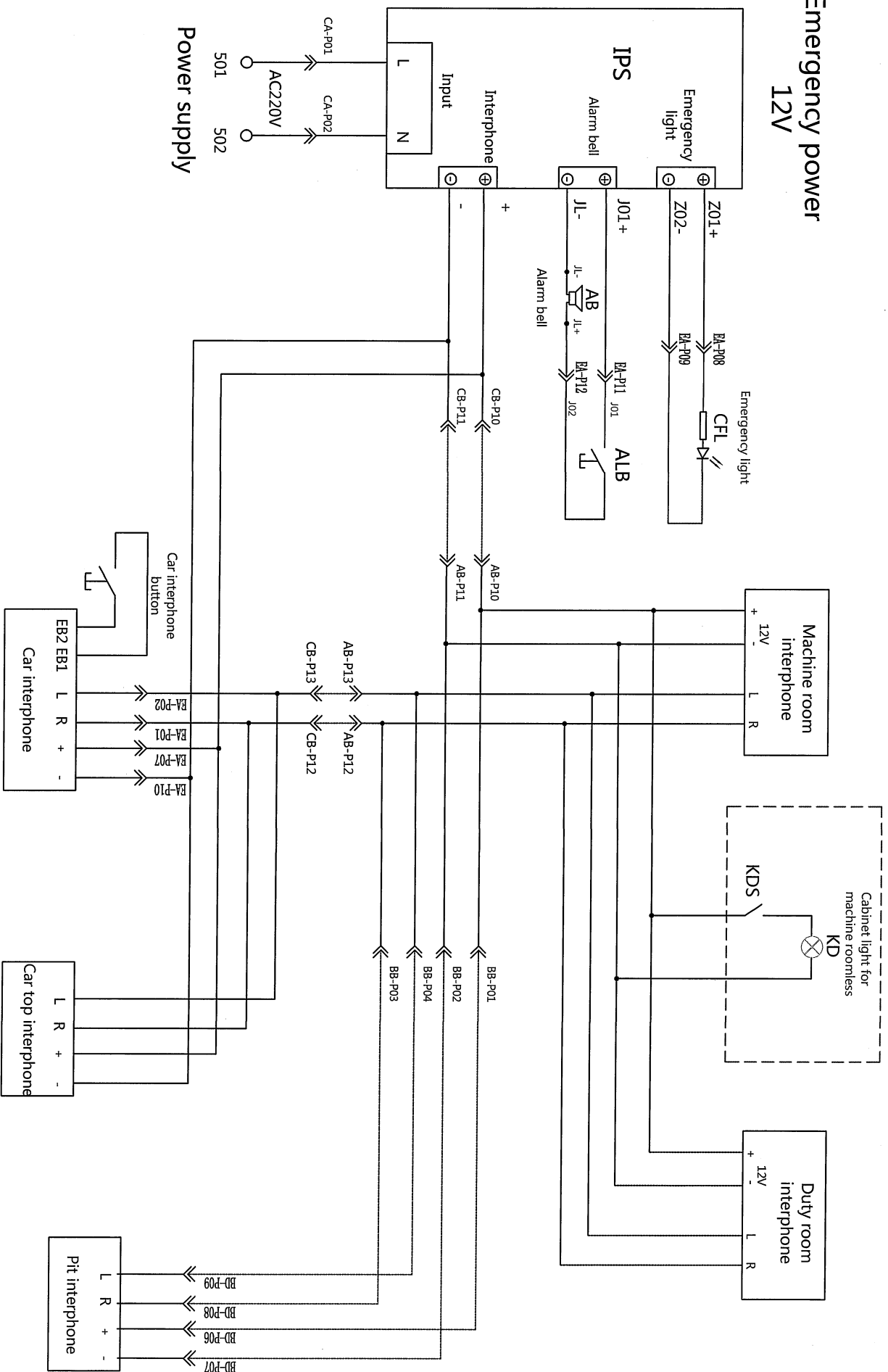
Lift 2



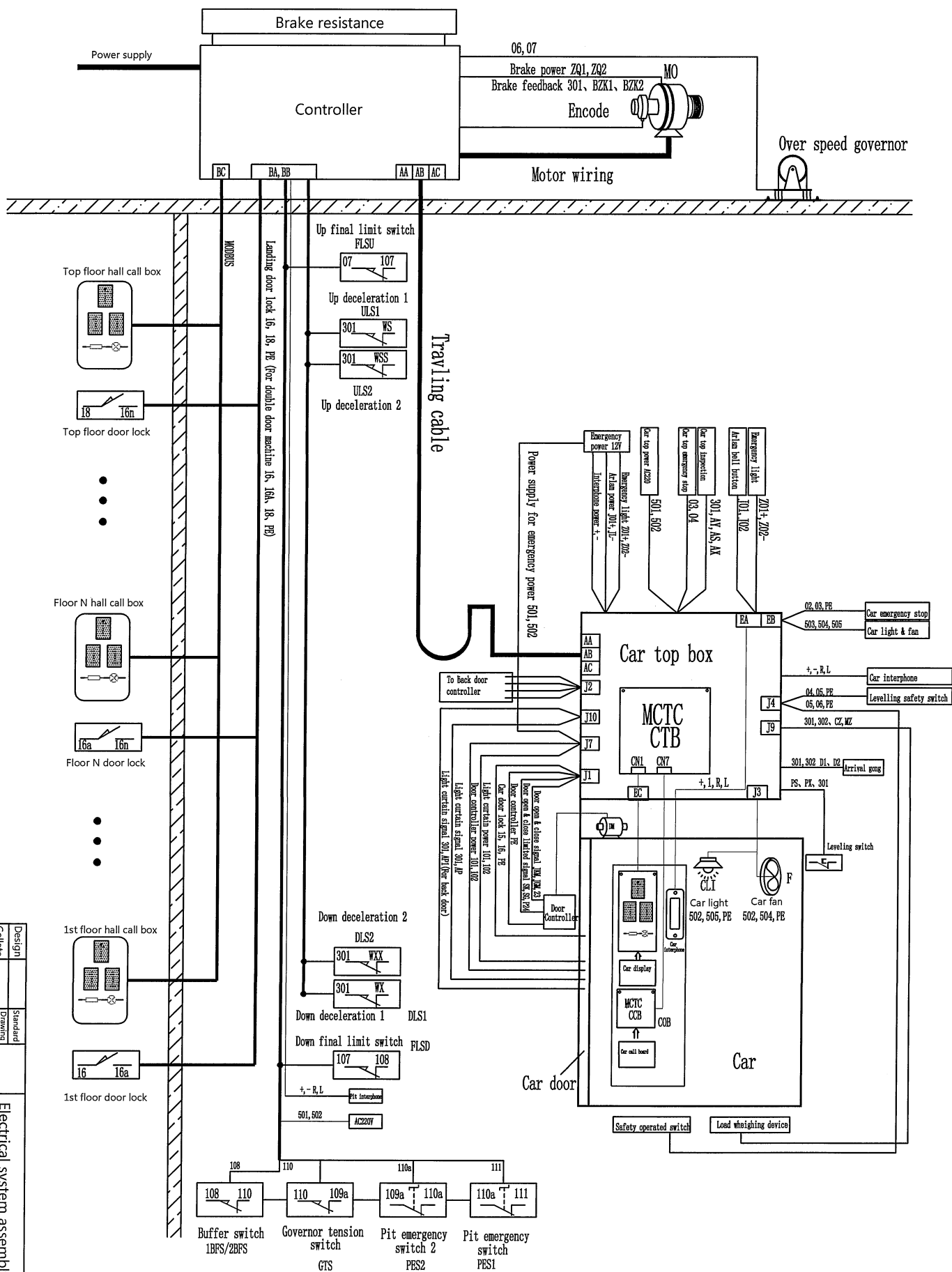
Notes:
 1、 The addresses of the HCBs should be set according to physical floors.
 2、 Parallel Control by Using CAN2 (terminal CN4), This mode can be implemented by directly connecting the CN4 terminals of two elevators and setting related parameters of group F6

Design	Standard	Parallel and hall show circuit	
Collate	Drawing NO.		
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Emergency power 12V

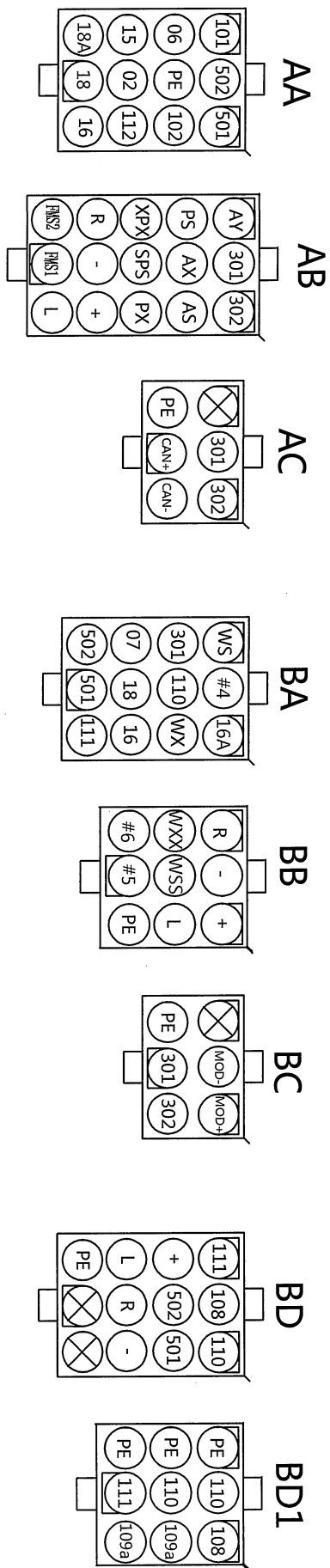


Design	Standard	Page 13	Intercom circuit
Cellate	Drawing NO.	V3.01	
Check	Versions		
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Design	Standard	Electrical system assembly drawing
Collate	Drawing	
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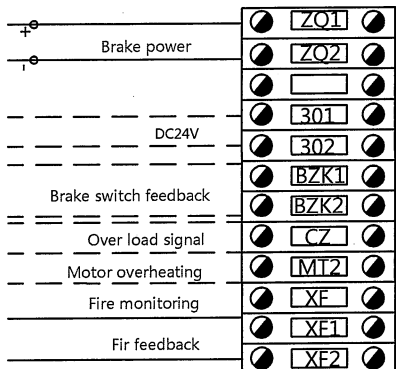


Connector of control cabinet traveling cable

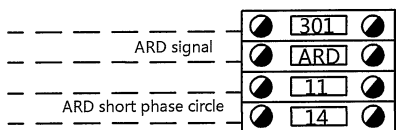
Connector of hoistway cable

Pit of hoistway cable

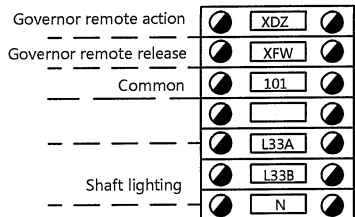
Terminal row TB



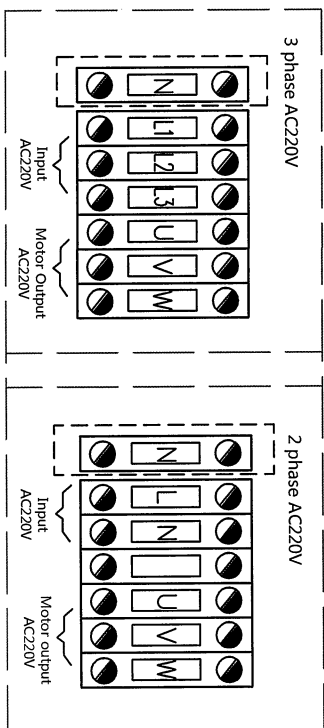
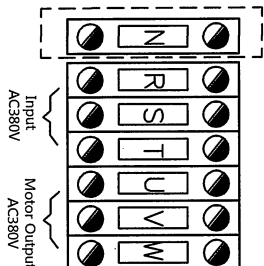
Applied in ARD case



When machine roomless, add

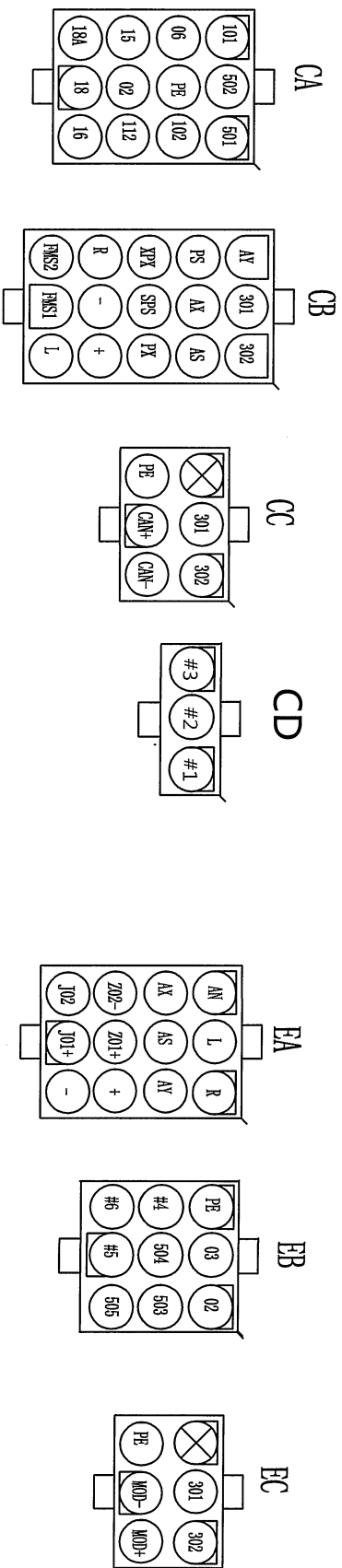


Terminal row TA



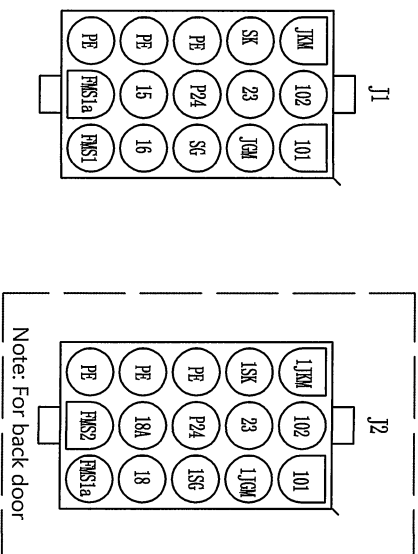
Machine room wiring

Design				Cabinet terminal block			
Collate	Standard	Drawing NO.		Page 16	NICE3000new		
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Connector of control cabinet traveling cable

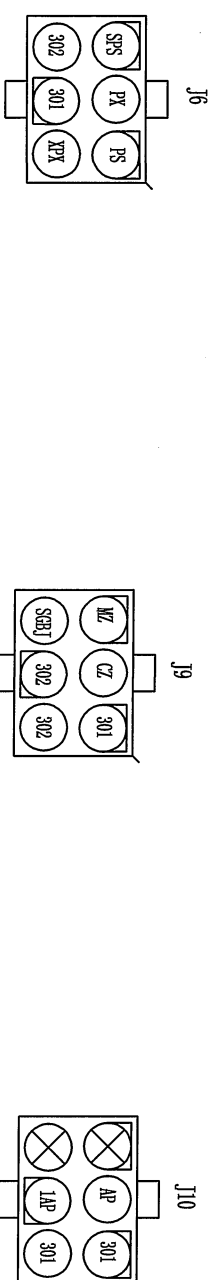
Connector of car operation box cable



Connector of door machine

Connector of Car top fan and light

Levelling safety and safety operated switch

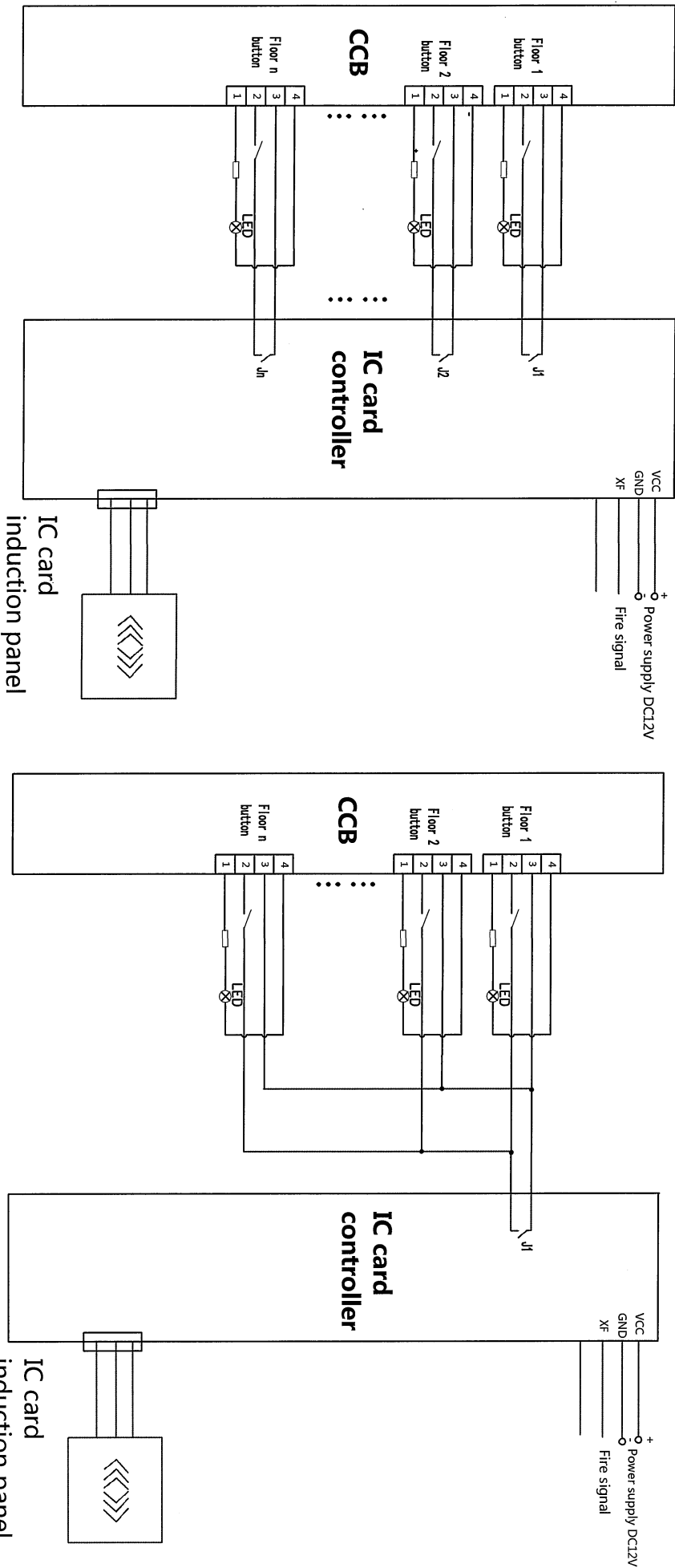


Leveling and re-level switch

Load weighing device

Light curtain

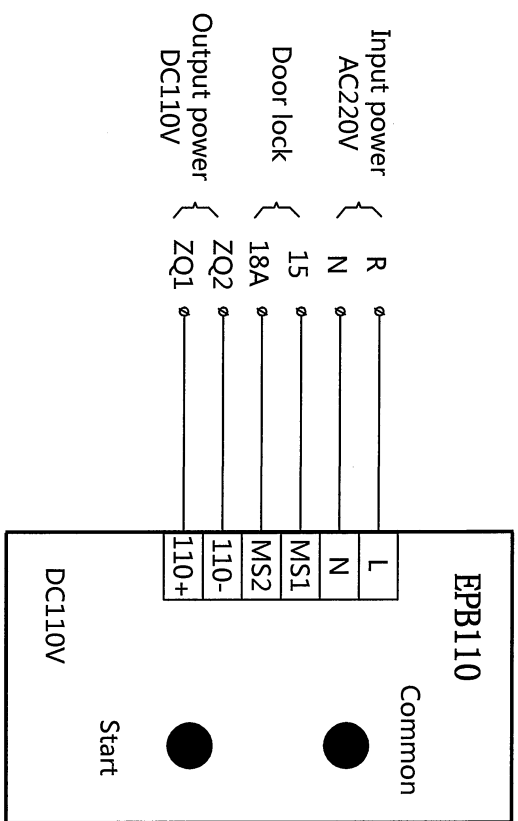
Design	Standard	Car top plug-in diagram	
Collate	Drawing NO.		
Check	Versions		
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		Total 20	



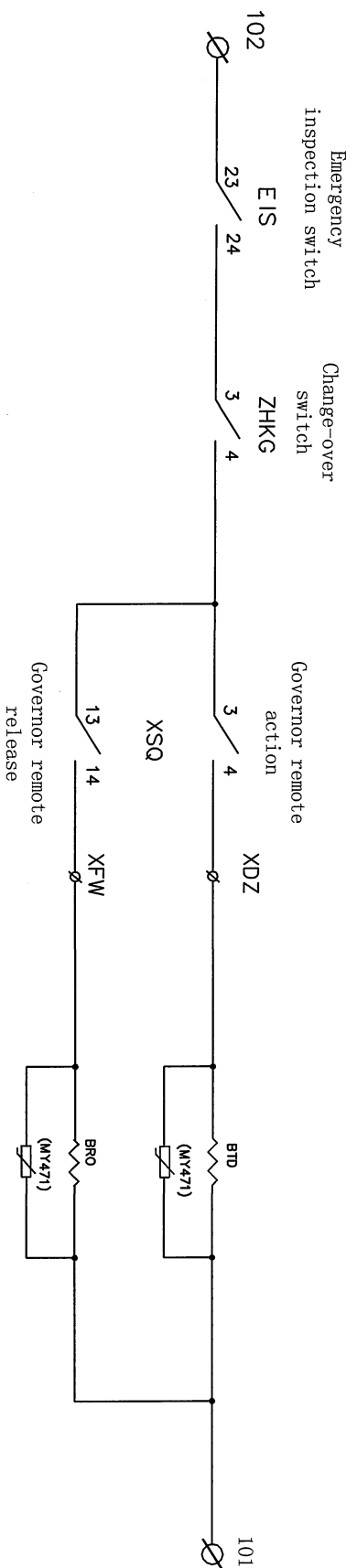
Design	Standard	IC card system control loop	
Collate	Drawing No.		
Check	V3.01	Page 18	NICE3000new
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Emergency brake-off instructions:

- 1, According to the diagram wiring, input AC220V, output DC110V;
- 2, The input power outage delay of 30 seconds before the loose operation;
- 3, Disconnect the total power supply of the control cabinet, confirm the car position, confirm that the car door lock and hall door lock are closed;
- 4, At the same time press and hold the "Common", "Start" button, DC110V output, the brake open, and the elevator car moves upward;
- 5, Observe the car move to the door area, release the brake button, the loose brake operation is completed.



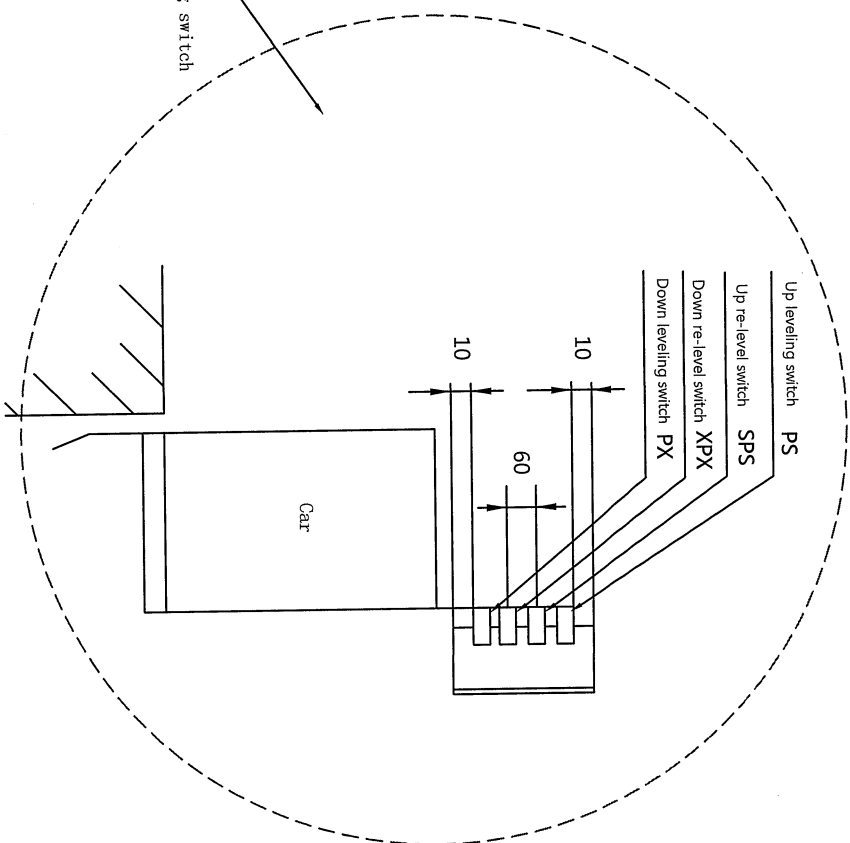
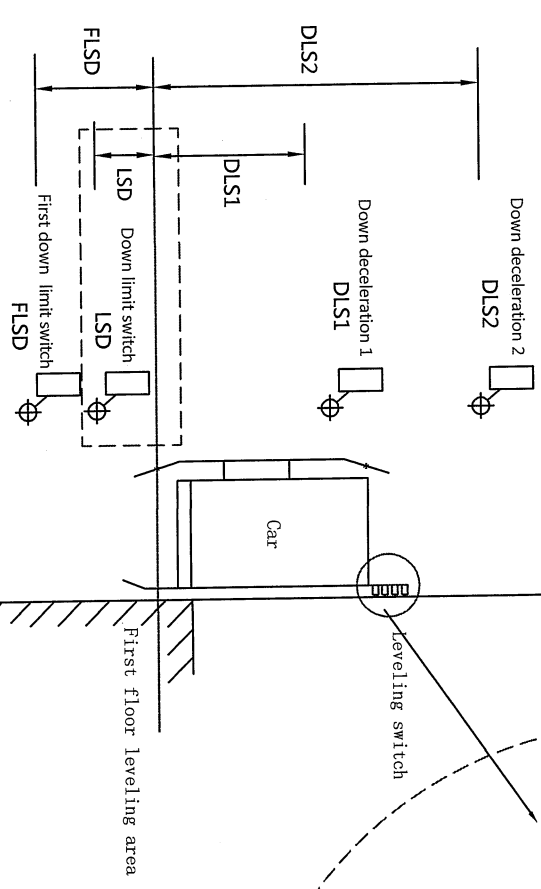
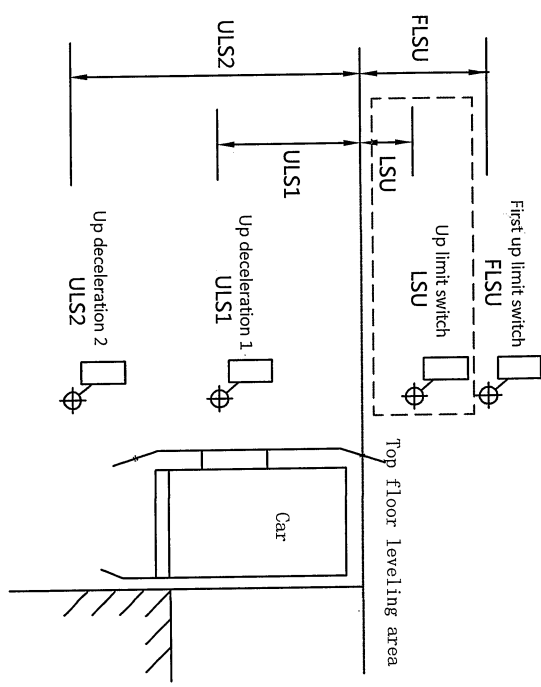
Backup power supply for brake



Note: Governor electromagnet power AC220V use 101, 102
Governor electromagnet power DC24v use 301, 302.

Design	Standard	Machine roomless governor circuit	
Collate	Drawing	Page 19	NICE3000new
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Switch	Distance		Speed									
			0.4m/s	0.5m/s	1.0m/s	1.5m/s	1.6m/s	1.75m/s	2.0m/s	2.5m/s		
First limit switch	FLSU,FLSD	0.15	0.15	0.15	0.15	0.15	0.15	0.15	0.18	0.18		
Unit switch	LSU,LSD	0.03~0.05	0.03~0.05	0.03~0.05	0.03~0.05	0.03~0.05	0.03~0.05	0.03~0.05	0.03~0.05	0.03~0.05		
Deceleration 1 switch	ULS1,DLS1	0.4	0.4	0.7	1.5	1.7	2.0	2.0	2.0	2.0		
Deceleration 2 switch	ULS2,DLS2								2.5	4.0		



Notes:
1, Soft limit function : P6-40 bit1=1,
2, When have UOMP function or Re-leveling
function or door pre-open function, use re-level
switch.

Design	Standard	Shaft position information	
Collate	Drawing NO.		
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